

NORTH ISLAND NAS, CA, USA

WMO: 722906

Lat: 32.700N

Lon: 117.200W

Elev: 8

StdP: 101.23

Time Zone: -8.00 (NAP)

Period: 94-19

WBAN: 93112

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	7.1	8.0	-6.0	2.3	18.3	-3.0	2.9	17.7	10.9	15.0	9.3	15.0	1.0	90	0.366

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	4.9	29.5	17.7	27.5	18.4	26.2	18.8	21.9	25.5	21.3	24.8	20.7	24.2	4.0	320

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
20.8	15.5	23.8	20.0	14.7	23.5	19.2	13.9	22.9	64.0	25.8	61.8	24.8	59.6	24.4	27.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.3	7.4	6.5	DB	4.6	34.2	1.1	3.0	3.8	36.4	3.2	38.2	2.6	39.9	1.8	42.1	(4)
(5)			WB	1.7	23.2	1.3	1.2	0.8	24.1	0.0	24.8	-0.7	25.5	-1.7	26.4	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	17.9	14.5	14.7	15.7	16.6	17.9	19.2	21.2	22.1	21.8	19.7	16.9	14.3	(6)	
	DBStd	3.48	2.48	2.26	2.23	2.26	1.99	1.65	1.72	1.72	2.21	2.19	2.58	2.11	(7)	
	HDD10.0	1	1	0	0	0	0	0	0	0	0	0	0	1	(8)	
	HDD18.3	607	121	104	88	62	31	8	1	0	1	8	57	126	(9)	
	CDD10.0	2882	141	132	176	197	244	276	346	374	353	302	207	134	(10)	
	CDD18.3	445	3	3	6	9	17	34	88	116	103	51	14	1	(11)	
	CDH23.3	1374	39	28	44	70	66	33	150	286	336	211	96	13	(12)	
(13)	CDH26.7	285	5	4	9	23	28	5	8	34	71	66	29	1	(13)	
(14)	Wind	WSAvg	2.8	2.4	2.9	3.0	3.3	3.2	3.1	3.0	3.0	2.6	2.3	2.4	(14)	
(15) Precipitation	PrecAvg	220	42	41	40	18	4	1	0	0	6	7	31	34	(15)	
	PrecMax	505	154	175	161	74	32	17	3	7	36	52	146	142	(16)	
	PrecMin	65	0	0	0	0	0	0	0	0	0	0	0	5	(17)	
	PrecStd	98	43	44	40	21	7	3	1	1	10	11	33	30	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	27.2	26.8	27.9	29.7	30.5	26.3	27.9	29.5	31.7	32.1	30.3	25.3	(19)	
		MCWB	14.1	13.7	14.2	15.1	15.0	19.6	21.1	20.9	21.2	16.5	14.8	12.1	(20)	
	2%	DB	23.9	23.2	23.7	24.7	23.8	23.9	26.2	27.4	28.4	28.2	26.0	22.1	(21)	
		MCWB	13.0	12.9	13.7	14.4	17.0	18.3	20.4	20.9	20.5	15.9	14.4	12.3	(22)	
	5%	DB	21.1	20.3	21.1	22.0	21.9	22.8	25.0	26.0	26.5	25.3	22.8	19.9	(23)	
		MCWB	12.7	13.1	14.0	14.9	16.5	17.7	19.9	20.5	20.6	17.1	14.2	13.0	(24)	
	10%	DB	18.7	18.3	19.1	20.0	20.6	21.9	23.9	24.8	25.1	23.2	20.9	18.3	(25)	
		MCWB	12.8	13.0	13.9	14.5	15.9	17.4	19.5	19.9	20.1	17.5	14.7	13.1	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	17.2	16.5	16.8	17.4	19.2	20.6	22.3	22.3	23.0	21.4	18.2	16.9	(27)	
		MCDB	19.3	19.3	20.8	21.5	22.9	24.0	25.8	26.2	27.7	24.7	21.1	18.8	(28)	
	2%	WB	15.7	15.3	15.8	16.5	18.0	19.4	21.2	21.6	22.1	20.0	17.3	15.8	(29)	
		MCDB	18.0	18.7	19.7	20.7	21.2	22.8	24.6	25.3	25.9	23.5	20.2	18.0	(30)	
	5%	WB	15.0	14.7	15.2	15.8	17.3	18.6	20.5	21.0	21.5	19.0	16.7	15.1	(31)	
		MCDB	17.6	17.9	18.8	19.7	20.3	21.7	23.9	24.7	25.0	22.4	19.8	17.5	(32)	
	10%	WB	14.2	14.1	14.7	15.2	16.6	17.8	20.0	20.5	20.8	18.3	16.0	14.3	(33)	
		MCDB	16.9	17.1	18.1	19.0	19.6	21.0	23.2	24.1	24.2	21.5	19.3	16.9	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	8.5	7.6	6.6	6.2	4.9	4.4	4.6	4.9	5.5	6.9	8.2	8.6	(35)	
		MCDBR	13.3	11.9	10.8	10.7	7.4	5.8	5.9	6.3	8.0	12.0	12.6	11.9	(36)	
	5% WB	MCWBR	5.9	5.5	4.9	4.8	3.4	2.7	2.5	2.6	3.2	5.3	5.9	6.2	(37)	
		MCWBR	7.9	7.6	7.2	7.7	6.3	5.9	5.6	5.9	7.0	7.9	8.4	7.4	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.316	0.327	0.340	0.351	0.371	0.370	0.399	0.397	0.364	0.370	0.324	0.327	(40)	
		taud	2.554	2.536	2.483	2.456	2.421	2.426	2.429	2.413	2.482	2.443	2.559	2.551	(41)	
	Ebn at Noon		908	931	939	937	915	911	883	882	902	869	891	874	(42)	
		Edh at Noon	82	92	104	112	117	116	115	115	103	100	81	79	(43)	
(44) All-Sky Solar Radiation	RadAvg		3.30	4.17	5.45	6.56	6.83	7.06	6.96	6.73	5.83	4.56	3.58	2.99	(44)	
	RadStd		0.20	0.34	0.31	0.37	0.53	0.46	0.48	0.30	0.33	0.34	0.25	0.21	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	+0.59	N/A	N/A	+1.07	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (38 neighbors)	+0.64	N/A	-1.59	+0.76	N/A	N/A	N/A	-105	+230	+119

Nomenclature: See separate page